Venkata Lakshmi Parimala Pasupuleti

AI + Data Engineer · Python · LangGraph · Agentic AI · LLM Evaluation · ETL/ELT Pipelines · FastAPI · AWS · Production Systems · M.S. Data Analytics Engineering

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PROFESSIONAL SUMMARY

Senior Software + AI/Data Engineer with 8 years owning production backend systems, ETL/ELT pipelines, and compliance-critical data workflows in enterprise SaaS. At Entrata (PropTech, 100K+ MAU): long-term owner of HUD income-limit import pipeline (180K domain records driving eligibility and rent calculations for 100K+ housing units), RabbitMQ-based async processing, 200+ SQL optimizations (30% gain), and AWS S3 document migration with 100% PHI compliance via Libsodium. Built **Leasel**, an 8-agent LangGraph agentic AI platform (93.6% precision, 87.9% F1 on 1,240 lease PDFs) with deterministic compliance checks, HITL escalation, and citation grounding. Built **AgentProbe**, an open-source AI evaluation toolkit with a 5-metric eval engine, CI-integrated regression harness, and multi-provider LLM routing (OpenAI / Anthropic / Groq / Ollama) - zero vendor dependency. Built a LangGraph mission-planning system for drone survey operations with FAA Part 107 RAG compliance layer. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

TECHNICAL SKILLS

Languages & Frameworks: Python (expert), PHP, JavaScript, TypeScript, Java, Kotlin, R, SQL (expert — schema design, execution plan tuning, 200+ optimizations), PL/SQL, Bash, Node.js, Scala; FastAPI, Django, Spring Boot, Laravel, React.js, Redux, Angular, GraphQL, REST APIs, OpenAPI

AI & Agentic Systems: LangGraph, LangChain, Multi-Agent Orchestration, Agentic AI, Tool Calling, MCP, RAG Pipelines, Vector Databases (ChromaDB, Qdrant, Pinecone, Weaviate), Hybrid Search, Semantic Search, Reranking, Contextual Retrieval, LLM Evaluation, LLM-as-Judge, HITL, Eval-Driven Development, Prompt Engineering, Few-Shot & Chain-of-Thought Prompting, Output Validation, Guardrails, Hallucination Detection, Agent Observability, Streaming Inference, OpenAI API, Anthropic API, Groq, Ollama, LangSmith, Gemini, AWS Bedrock, Azure OpenAI Service

ML & Data Science: PyTorch, TensorFlow, scikit-learn, XGBoost, Random Forest, Isolation Forest, Logistic Regression, Anomaly Detection, Fraud Detection, NLP, BERT, HuggingFace Transformers, Model Fine-Tuning, spaCy, NLTK, Feature Engineering, Statistical Modeling, Hypothesis Testing, Pandas, NumPy, Matplotlib, Seaborn, Plotly, SciPy, MLflow, Jupyter

Cloud & Infrastructure: AWS (S3, EC2, RDS, Glue, Lambda, SageMaker, Bedrock, Step Functions, SQS, OpenSearch, API Gateway, IAM), Azure (Data Factory, Data Lake, Databricks, ML, AI Foundry, AI Search, OpenAI Service, Cognitive Services, DevOps), GCP / Vertex AI, Docker, Kubernetes, CI/CD (GitHub Actions), Terraform, Prometheus, Grafana, OpenTelemetry, Apache Kafka, RabbitMQ, Celery, Apache Airflow, Redis, Apache Spark, PySpark

Data Engineering & Databases: ETL/ELT Pipelines, Schema Design & Normalization, Dimensional Modeling, Medallion Architecture, Lakehouse, Star/Snowflake Schema, Data Lineage, Data Governance, Data Quality, PostgreSQL, MySQL, SQLite, Snowflake, Oracle, Microsoft SQL Server (T-SQL), Teradata, Solr, NoSQL, Tableau, Power BI, ggplot2, Quarto

Enterprise & Security: Salesforce, Microsoft Power Apps, Dell Boomi, Third-Party Integrations, Libsodium Encryption, PHI / HIPAA Compliance, OWASP LLM Top 10, Audit Logging, SOC 2, NIST, Microservices, Multi-Tenant Architecture, Agile/Scrum (120+ sprints), Git, Linux, Jira

WORK EXPERIENCE

Blue Horse Digital
Sep 2026 – Present · Virginia, USA

AI/ML Engineer (Volunteer) — ML Pipelines · Business Intelligence · Data-Driven Features

- Building ML models and data analysis pipelines for BI projects — data collection, cleaning, feature engineering, model training, and evaluation; integrating AI components with back-end and front-end teams; surfacing insights from raw business data into application-facing APIs and reporting layers.

George Mason University — College of Engineering and Computing
Aug 2025 – May 2026 · Part-time · On-site

Graduate Teaching Assistant — Python Programming · Database Design · MIS 303

- Delivered technical workshops to **600+ students** across multiple sections covering Python (data structures, OOP, automation scripting) and database design (ER modeling, SQL, MS Access/Excel); provided one-on-one debugging support and graded assignments with rubric-aligned feedback.

Magizhs Tech Solution
May 2025 – Aug 2025

AI/ML Engineer Intern — LangGraph · Agentic Systems · RAG · Drone Mission Planning

- Built a LangGraph stateful agentic mission-planning system (parse → plan → retrieve → validate → conditional routing) for drone survey operations — parsed natural-language requests from marketing, real estate, and commercial stakeholders into structured flight plans using photogrammetry math (GSD, camera footprint, overlap-driven line spacing).
- Implemented RAG layer over FAA Part 107 corpus for compliance Q&A with cited sources; engineered guardrails enforcing altitude limits, battery budgets, and geometric no-fly-zone checks before MAVLink/QGroundControl export.

George Mason University
Aug 2024 – Aug 2025 · Virginia, USA

Research Associate — ML Modeling · ETL Pipelines · Analytics Engineering · Model Fine-Tuning

- Built multi-source prediction models (Python, R) on WHO and UK Road Safety datasets across 190+ countries — **$R^2 = 0.977$** regression accuracy; applied Isolation Forest to automate data integrity checks across 3 research pipelines, surfacing anomalies missed by manual review.
- Fine-tuned a custom classification model (HuggingFace Transformers + PyTorch); deployed inference API on Quart (async Python); designed Tableau, ggplot2, and Quarto dashboards for faculty leadership — **40% improvement** in research comprehension across two cohorts.

Entrata — PropTech / Real Estate SaaS (Lehi, UT · Remote)

Sep 2018 – Aug 2024

Senior Software Engineer — HUD Compliance · ETL Pipelines · Async Processing · AWS · Production AI

- **Owned the annual HUD income-limit import pipeline for nearly 7 years** — a few thousand federal source rows expanded into ~180,000 domain records powering eligibility, rent, HAP, and tax-credit calculations for 100K+ housing units; zero compliance violations across the ownership period.
- **Built and maintained RabbitMQ-based async processing** for gross-rent updates across thousands of tenant records — retry logic and idempotency guarantees eliminated double-processing failures; owned the system through production incidents and annual rent-schedule changes.
- **Optimized 200+ SQL queries (30% performance gain)** via execution plan analysis, indexing, partitioning, and stored procedure refactoring; owned complex schemas for eligibility, financial, and document workflows across multi-tenant production environments.
- **Led AWS S3 document migration** for the Affordable Housing module — 100% PHI compliance with Libsodium-encrypted pipelines; zero security incidents across the entire resident data migration.
- **Shipped 20+ full-stack production modules** (PHP, Python, PostgreSQL, JavaScript, RabbitMQ) across 120+ Agile sprints; led microservices redesign with Docker (50% scalability improvement, 30% deployment reduction); eliminated 80% of manual operational workload via Python/Bash automation.
- Mentored 5+ junior engineers; conducted **300+ peer code reviews**; built internal debugging tools that reduced debugging time by 40% and kept bug count below 10 for 6+ consecutive months.

Entrata India (Db Xento Systems) — Pune

Jan 2017 – Sep 2018

Associate / Trainee Software Engineer — Full-Stack · Data Pipelines · AWS S3 · Schema Design

- Built property management modules (PHP, Python, Java, PostgreSQL) for affordable housing; designed SQL import pipelines for 1.6M HUD/Tax Credit records; led cloud migration from legacy server storage to AWS S3 (resolved 90% of legacy space constraints); created 56 state-specific housing certification documents with merge fields; redesigned document storage database architecture, migrating hundreds of thousands of records to a modern schema.

PROJECTS

Leasel: Agentic AI Support Investigation Platform

PROPTECH · LLM EVALUATION · AGENTIC AI

Python · LangGraph (8-agent) · FastAPI · ChromaDB · TF-IDF · OCR · SQLite · LLM-as-Judge · HITL — github.com/venkatalakshmiparimala/Lease_complaince_rag

- Three-surface enterprise AI knowledge platform sharing one evidence model: **Document Intelligence** (OCR-fallback ingestion, section-aware chunking, hybrid TF-IDF + semantic reranking, citation grounding, missing/conflicting clause detection, HITL gate on regulated decisions); **Support Investigation** (8-agent LangGraph pipeline: Triage → Retrieval → Configuration → Historical Incident → Structured Data → Reasoning → Validation → Human Escalation, producing ranked evidence-backed root-cause hypotheses); **Tenant Self-Service** (policy Q&A with hard rule: eligibility decisions always route to staff, never resolved by AI).
- **93.6% precision, 87.9% F1** across 10 compliance rules on 1,240 residential lease PDFs (5 US states + federal, 1,050 synthetic tenants); pass@1 81.7%, pass@3 88.3% via multi-trial sampling with regression gate and LLM-as-judge scoring.

AgentProbe: Open-Source AI Evaluation Toolkit

LLM EVALUATION · AI-ASSISTED DEV

Python · FastAPI · LangGraph · ChromaDB · React · CI/CD (GitHub Actions) · OpenAI / Anthropic / Groq / Ollama — github.com/venkatalakshmiparimala/AgentProbe

- Fully local, zero-cost AI agent evaluation toolkit: 15-endpoint FastAPI backend, **5-metric eval engine** (correctness, groundedness, latency, token efficiency, hallucination rate), CI-integrated regression harness with golden datasets, multi-provider LLM routing (OpenAI / Anthropic / Groq / Ollama), React observability dashboard tracking per-run metrics and retrieval drift, offline + online evaluation loops — used to evaluate Leasel across 1,240+ compliance test cases.

AI on FHIR: NLP-Driven Medical Records Query System

CLINICAL AI

Flask · React · LLMs · BERT · Embeddings · spaCy · FHIR APIs · Docker — github.com/venkatalakshmiparimala/AI_on_FHIR

- Reduced manual clinical retrieval time ~60% by translating natural-language queries (EN/ES) into FHIR API calls using LLMs, BERT, and embeddings; enforced HIPAA-compliant RBAC across structured medical records.

CrashClarity: Large-Scale Road Accident Analysis (104K Records)

ANALYTICS ENGINEERING

Python · R · scikit-learn · Random Forest · Logistic Regression · Feature Engineering · ETL · Tableau · GitHub Pages — Portfolio

- End-to-end analysis of 104K+ UK accident records — Random Forest + Logistic Regression pipeline with feature engineering across weather, road, and time variables (**85%+ accuracy**); deployed interactive dashboard on GitHub Pages.

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia

May 2026 · GPA 3.87

Data engineering, ML systems, statistical modeling, AI pipelines, data governance, anomaly detection, NLP, computer vision

B.Tech Computer Science & Engineering — JNTU Anantapur

2013 – 2017